SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofArtificialIntelligenceandDataScience

ASSESSMENTTOOL1 –ANALYTICALPROBLEMSOLVING

CourseCode:	DSA03	CourseTitle:	NaturalLanguageProcessing
COAssessed:	CO2 – Manipulation	Assessment:	AssessmentTool1–ANALYTICALPRO SOLVING
Weightage:	50% of		
CO5Statement:	Designinflectional,derivational,finite-stateparsing,stemmingalgorithmsandforanaly semorphological methods. (BL-3)		
Student Name:	D.Chinna Moulali	Reg. No.:	19242
Section/Batch:	B.TechAI&ML	Date:	29-07-2026

CodeofConduct

I D.Chinna Moulali (Name /Reg No) certify that thissubmission is myoriginalworkand thatIhaveadheredtotheguidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrity rules will result in disciplinary action.

Signature of theStudent:

Instructions

- Readallquestionscarefullybeforeattemptingtheanswers.
- Answerallfivequestions.Eachquestioncarriesequalmarks.
- Showallintermediatesteps,calculations,probabilitycomputations,taggingdecisions,andjustifications wherever applicable.
- Duration:30minutes.

EnglishWordClasses,Rule-BasedPartofSpe Tagging, Counting Words	Porter Stemme ofrelational,rela fixremovaland stem extraction

EnglishWordClasses,PartofSpeech	Morphologicalc and connecti suffixes,inflec ionfordocume
EnglishWordClasses,TagsetsforEnglish,Rul sed Part of Speech Tagging	Morphologica happily; ident semantic effe
CountingWords,PartofSpeechTaggi ng,English Word Classes	Stemming ar handling grar search prepro
Finite-StateMorphologicalAnalysis,Rule-Bas ed Part of Speech Tagging, Transformation-Based Tagging	Finite-state p modelingregu inflectionsnor

Q1.MorphologicalAnalysisPipeline

InputWords:

- Connected
- Connecting
- Connection

Step1:DecomposeintoRootandSuffix

Word	R	
Connected		
Connecting		
Connection		

Step2:Classification

- Connected→Inflectional(Pasttense)
- Connecting→Inflectional(Presentparticiple)
- Connection→Derivational(Formsanoun)

Step3:Normalization

Allwordsareconvertedtothecommonbaseform:

Connect

ParsedStructure

- Connected=Connect+ ed
- Connecting=Connect+ ing
- Connection=Connect+ ion

FinalOutput

NormalizedRoot=Connect

Q2.MorphologicalParsingModule

InputWords

- Unhappy
- Happiness

- Happily

Step1:MorphologicalBreakdown
Suffix
—

Happiness			—
Happily			

Step2:Classification

Word	
Unhappy	

- Unhappy→Derivational
- Happiness→Derivational
- Happily→Derivational

SemanticEffect

- un-changesmeaningtoopposite.
- -nesschanges adjectiveintonoun.
- -lychanges adjectiveintoadverb.

FinalRoot
Happy

Q3.Stemming-BasedPreprocessing

InputWords

- Played
- Player
- Playing

Step1:RemoveAffixes

Word
Played
Player
Playing

Affix
-ed
-er
-ing

Step2:Classification

- Played→Inflectional
- Player→Derivational
- Playing → Inflectional

FinalStem Play

Result
Allwordsareindexed as:
Play

Q4.Finite-StateMorphologicalParsing

InputWords

- Writes
- Writing
- Written

MorphologicalBreakdown

Word		Root
Writes		W
Writing		
Written		

InputWords

- Relational
- Relation
- Relate

Step-by-StepStemming

Word	RemoveSu	Interme
Relational	-ational	Relate
Relation		Rela
Relate		Re

StateTransitions

StartState→ Root (Write)

- Write→ +s→ Writes
- Write→ +ing→ Writing
- Write→+en→Written

NormalizedRoot

Write

Result

Allformsmap to:

Write

Explanation

- Removederivationalsuffixes.
- ApplyPorterStemmerrules.
- Convertallwordstoone commonstem.

Q5.PorterStemmer-BasedProcessing

FinalStem

Relat

Result

Allwordsareindexed as:

Relat

FinalStem

Conclusion

Morphologicalanalysisandstemminghelpconvertdifferentformsofawordintoacommonrootorstem. This improves document indexing, searching, and information retrieval by treatingrelatedwords as the same term.

RubricsforCalculation

NumericalProblems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1(Beginning)

Problem Understa nding	20%	Clearly interpretsproblem,i sallparametersand requirements accurately	Understands problemwithmi norgaps	Partial understanding; misses elements	Misinterprets unableto understand key problem
Method Selection	20%	Selects appropriate method/formul a withjustificatio	most Select methodwithminor errors strong	Inappropriateor partially method	Unableto correct selectmethod
Solution Process	25%	Logical,systematic withapproach throughout	Mostly correct stepswithminor mistakes	correct Disorganizedste ps; multiple	Noclear process errors
Accuracy of Calculatio n	20%	All calculations accurate;correct final answer	Minorcalculation errors	Frequenterrors; incorrect fina answer	No calculations
Interpreta tion Results	15%	Clearlyinterprets resultswithunits and engineering relevance	Basic interpretationwit minor omissions	Limitedorunclear interpretation	No interpretation

valid

Q.No./ Criteria Level	Problem Understanding	Method Selection	Solution Process
1			
2			
3			
4			
5			

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature:	Signature:	Signature:
Date:	Date:	Date: